W-F-414B
February 9, 1973
SUPERSEDING
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(GSA-FSS) March 25, 1971 and
W-F-414a May 8, 1957

FEDERAL SPECIFICATION

FIXTURE, LIGHTING (FLUORESCENT, ALTERNATING-CURRENT, PENDANT MOUNTING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pendant mounted, fluorescent lamp lighting fixtures for alternating current service, as specified (see 3.18). Unless otherwise specified, lamps and starters shall not be included with the fixtures.

1.2 Classification. Lighting fixtures furnished under this specification shall be of the following types, styles, mountings, and lamp arrangements, as specified (see 6.1):

Types:

Type I - Lighting fixtures which have been specifically designed to permit interchangeability of parts of different manufacturers (see 3.2).

Type II - General Purposes Lighting Fixtures.

Styles:

Style A - Direct.

Style B - Semi-direct.

Style C - Direct-indirect.

Style D - Semi-indirect.

Mountings:

Mounting 1 - Individual.

Mounting 2 - Continuous Row.

Lamp Arrangements:

Arrangement A - Two 40-watt, medium bi-pin base, T-12 rapid-start, fluorescent lamps in accordance with ANSI Standard C78.700.

Arrangement B - Four 40-watt, medium bi-pin base, T-12 rapid-start, fluorescent lamps in accordance with ANSI Standard C78.700.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

NN-P-71 - Pallets, Material Handling, Wood, Double Faced, Stringer Construction.
W-B-30 - Ballast, Fluorescent Lamp.
QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood Cleated-Plywood.
PPP-B-621 - Boxes, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fireboard, Corrugated, Triple-wall.
PPP-B-665 - Boxes, Paperboard, Metal-Stayed.
PPP-B-676 - Boxes, Setup.
PPP-S-760 - Strapping, Nonmetallic.
PPP-T-97 - Tape, pressure-sensitive adhesive, filament reinforced.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection Sampling and Testing.
Fed. Test Method Std. No. 151 - Metals: Test Methods.
Fed. Test Method Std. No. 406 - Plastics; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activites may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Methods of Preservation.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-463 - Definitions and System of Units, Electromagnetic Interference Technology.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents from a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal shall apply:

General Services Administration:

Public Building Service Standard Details Nos. 1640-2, 1640-6, and 1640-7.
Public Building Service Standard Lighting Fixture Specification.

(Copies may be obtained from General Services Administration, Public Buildings Service, Washington, DC 20405.)

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Underwriters' Laboratories (UL) Inc. Standards:

No. 57 - Electric Lighting Fixtures.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

American National Standards Institute (ANSI), Inc. Standard:

C78.700 - 400-WATT T-12 Rapid-Start Fluorescent Lamp.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, N.Y. 10018.)

American Society for Testing and Materials (ASTM) Standard:

D788 - Standard Specification for Methacrylate Moldings and Extrusion Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Illuminating Engineering Society (IES) Handbook:

IES Lighting Handbook, Fifth Edition 1972.
RQQ Report No. 2 (1972) - Outline of a Standard Procedure for Computing Visual Comfort Ratings for Interior Lighting.
Journal of IES, Volume 1, No. 2, January, 1972.

(Application for copies should be addressed to the Illuminating Engineering Society, 345 East 47th Street, New York, N.Y. 10017.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Fire, Safety and Casualty Hazards.

3.1.1 Each contractor shall submit to the contracting agency proof that each fixture he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc., (UL) Standard No. 57 - Electric Lighting Fixtures. Listing by the Underwriters' Laboratories, Inc. will be accepted as evidence that the product conforms to this requirement. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.2 Design.

3.2.1 Type I. Type I lighting fixtures shall be in accordance with the latest issue of the General Services Administration, Public Buildings Service Standard Lighting Fixture Specification and Public Buildings Service Standard Detail for the following styles, mountings and lamp arrangements listed in table I.

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TABLE I.

Style	Mountings	Lamp Arrangements	Detail Drawing Type No.	Standard Detail Drawing No.
C	1	A	363 RS	1640-6
C	1 and 2	A	362 RS	1640-6 and 1640-2
C	1 and 2	B	364 RS	1640-7 and 1640-2
C	1	B	365 RS	1640-7

3.2.1.1 Stems. Stems shall be as indicated on the standard detail drawings.

3.2.2 Type II. Unless otherwise specified (see 6.1), type II fixtures shall utilize flat bottom closures of the louvered or lens type. The lens type shall be of methyl-methacrylate plastic (see 3.4) or annealed glass smooth on one side with a uniform overall prismatic pattern on the other side. The overall thickness of lens type closures shall be a norminal 0.156 inch, minimum overall thickness shall be 0.151 inch. The louvered type closure shall be of aluminum, methyl-methacrylate plastic (see 3.4), or steel. Bottom closures shall be designed to preclude accidental (unintentional) disassembly from fixture housing and shall be of a positive release type. Touch type (pressure release) latches shall not be used. Additionally, hinged type bottom closures shall be of a design that precludes accidental separation from fixture housing when closure is in an open position, i.e., hanging by its hinges.

3.2.2.1 Canopy assembly and stems. Canopy assemblies shall be of a shallow design which will project not more than 2-1/4 inches from the ceiling. Additionally, all canopy assemblies shall be designed to completely cover a 4 inch square electrical outlet box. A ball and socket joint or other movable joint that allows the stem to move freely at least 10 degrees from the vertical in all directions when the fixture is installed, shall be contained within and must be completely covered by the canopy. At least two stems shall be provided with each fixture. Unless otherwise specified (see 6.1), stems shall be of a length that will permit the fixture body to hang 18 inches below ceiling, as measured from the ceiling surface to the lowest point on the fixture body when installed. For fixture leveling purposes, tems shall be of a design that permits the fixture body, when installed, to be raised or lowered at least one inch.

3.2.2.2 End plates. Lighting fixtures shall be provided with the following number of end plates, end caps, or end closures as required for the mounting specified:

2 for individual fixtures (Mounting 1).

1 for each fixture in a continuous row (Mounting 2).

3.2.2.3 Accessibility of parts. Lamps shall be easily removable and replaceable without the use of tools and without the removal of other lamps. This is not intended to preclude the use of hinged louvers. The ballast(s) and wiring shall be easily accessible without the use of special purpose tools. Also, ballast(s) and lampholders must be installed in a manner that will allow replacement without removal of the fixture from the ceiling.

3.3 Assembly and construction. The fixtures shall be completely assembled and ready for installation to the building wiring, except that bottom closures, canopy assemblies, coupler(s), stems and related installation hardware may be shipped unassembled. Type II fixtures shall be of a design that precludes an extension of panels, plates or any other type of baffle more than one inch below the bottom surface of the bottom closure. This requirement shall apply whether or not the extension is an integral part of housing or is an add on feature. Additionally, housing of type II fixtures shall be provided with a separate factory installed grounding device. The grounding device is to be used for connecting a separate grounding conductor to the fixture housing and for no other purpose. The device shall be a 10/32 machine screw threaded stud and nut, with a flat washer. The stud shall be either a welded or pressure fastened type and shall meet all applicable requirements of the National Electrical Code, latest edition. Except as otherwise noted, fixtures shall consist of all parts necessary to constitute a complete functional luminaire.

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3.3.1 Material. The material shall be free from all defects and imperfections which might affect the serviceability and appearance of the finished product.

3.3.2 Type I fixtures. Type I fixtures shall be constructed as specified on the standard detail drawing.

3.3.3 Type II fixtures. The construction of type II fixtures shall conform to the applicable requirements of UL-57.

3.4 Plastic and Glass Material.

3.4.1 Type I. Plastic side panels on type 1 fixtures shall be smooth, translucent, and of the thickness, size, and contour indicated on the standard detail drawings. The plastic shall be heat resistant 100 percent virgin methyl-methacrylate meeting the requirements specified in 3.4.2. Test results indicating compliance with ASTM D788 shall be required (see 4.5.1).

3.4.2 Type II. Plastic for bottom lens closure(s) and side panels (if used) shall be 100 percent virgin methyl-methacrylate in accordance with ASTM D788. Additionally, the plastic shall be either nonburning, self-extinguishing, or slow burning. The rate of burning shall not exceed 3.2 inches per minute for a 1/2 inch width of plastic when tested in accordance with 4.5.1.1. Test results indicating compliance with ASTM D788 shall be required(see 4.5.1). Glass panels (if used)shall have all edges "swiped" to remove sharpness.

3.4.3 Marking. Each type II lens closure and panel (if used) shall be permanently marked with the following identification. The identification shall include the lens manufacturer's name or trademark, coding or lens number and date (month and year of manufacture).

3.5 Lampholders. The lampholders shall be medium bi-pin type. Lampholders shall hold the lamps securely without the use of auxiliary devices. Lampholders shall be securely fastened to the mounting surface with necessary provisions to prevent the lampholders from turning. The lampholder and mounting shall be of a design that prevents spreading of the lampholders upon installation of the lamp(s). The longitudinal distance between the inner surface of a pair of lampholders shall be sufficient, but not greater than necessary to insure proper fit and contact of the lamp specified in ANSI C78.700.

3.5.1 Lamp spacing. For lamp arrangement A, the lamps shall be equidistant from the center of the fixture; for lamp arrangement B, the two outside lamps shall be equidistant from the center of the fixture and also the two inside lamps shall be equidistant from the center of the fixture. All lamps shall be spaced in a manner that will permit easy removal and insertion of lamps by hand without disturbing the remaining lamps.

3.6 Ballast. Ballasts shall conform to Federal Specification W-B-30 and shall be designed for operation on either 110-125 volt or 250-290 volt, 60-Hertz alternating current; or on 210-236 volt, 50-Hertz alternating current, as required (see 3.18 and 6.1). Additionally, ballasts shall be of the rapid start, high power factor type. The ballasts shall have approved built-in protective and automatic reset features. The built-in protective and automatic reset features shall be of the thermostatic type class "P", meeting UL requirements. The ballasts shall be held firmly in contact with the metal of the fixtureby 4 thread cutting (TC) sheet metal screws or by 4-bolts secured in a manner that precludes the bolts from turning.

3.7 Bolts, lockwashers, screws and washers. Bolts shall have either two nuts or one nut and a lockwasher. Flat washers shall be used where a bolt, screw, nut or lockwasher may dig into plastic.

3.8 Enclosures or chassis and mounting parts.

3.8.1 General. Type I enclosure or chassis shall be as indicated on the standard detail drawings. Ballasts may be attached to the raceway. All type II enclosures shall be fabricated in accordance with UL-57. Additionally, type II fixtures shall be constructed of either 20 gauge steel or 22 gauge steel with stiffening ribs. All necessary parts required for mounting the fixture to the ceiling shall be provided with each fixture.

3.8.2 Mounting l. Type II fixtures shall be provided with a suitable means for mounting. The top of the chassis shall have four knockouts to facilitate attachment of suspension stems. Two knockouts shall be located near the center of the chassis and one knockout near each end. All knockouts shall be provided for attaching the suspension stems. On type I fixtures the chassis shall be as indicated on the standard detail drawings.

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3.8.3 Mounting 2. These fixtures shall conform to the requirements for mounting 1. Also channels shall be so fabricated that an open continuous wireway will be formed when two or more fixtures are joined together to form a continuous row. Additionally, each fixture shall be provided with a coupler(s) for purposes of alignment and electrical continuity.

3.9 Marking. Unless otherwise specified (see 6.1), each fixture shall be marked in accordance with provisions of UL-57. Marking shall be discernible when fixture is assembled but shall not be visible from the floor. Additionally, the marking shall include FSN number, contract or order number, date (month and year) of manufacture and model number.

3.10 Finish on metal parts (type II fixtures).

3.10.1 Reflecting surface. Unless otherwise specified (see 6.1), the reflecting surface finish shall be of a pigmented synthetic enamel having a dry film thickness so as to completely hide a black and white substrate (see 4.5.3). Additionally, finish shall show no indication of chipping, cracking, flaking or any other sign of loss of adhesion when tested as specified in 4.5.3.

3.10.1.1 Reflectance. The average reflectance shall not decrease more than 4 percent from the value provided in independent laboratory test data (see 4.6.5) when tested in accordance with paragraph 4.5.3. Also, the average fixture reflectance when determined in accordance 4.5.4 shall be at least 96 percent of the value given in independent test laboratory data.

3.10.2 Nonreflecting surfaces. Unless otherwise specified (see 6.1), the finish on all nonreflecting exterior surfaces shall be aluminum oxide on aluminum; white, gray, or aluminum paint on steel; nickel or chromium plating on copper alloy. Fastening devices shall be nickel, chromium, cadmium, or zinc plated. All painted surfaces shall be free from tears, star marks, crazing, blisters, black specks, hairlines, chipping, pinholes and other defects which may impair appearance or serviceability.

3.11 Salt spray test. When specified (see 6.1), one complete fixture, less the electrical components, shall be subjected to the salt spray test of paragraph 4.5.2. Upon completion of the test there shall be no evidence of corrosion.

3.12 Fixture rigidity (type II fixtures only). The assembled fixture with required number of lamps inserted therein, shall be subjected to the test specified in 4.5.5. Testing shall be accomplished with and without bottom closure(s) installed. When tested in the manner outlined, neither fixture nor lamps shall become damaged or disassembled.

3.13 Interchangeability of parts. The parts of each type II fixture of a particular model number shall be interchangeable with any of the fixtures of the same model number produced by the same manufacturer. The parts of type I fixtures shall be interchangeable with the fixtures of other manufacturers.

3.14 Light distribution (type II fixtures only). When determined in accordance with paragraph 4.6.1, the percent of light distribution, expressed as a percentage of the lighting fixture's total output in lumens for the style of fixture specified in paragraph 1.2, shall be as follows:

Style A - 0-10 percent upward 100-90 percent downward.

Style B - 10-40 percent upward 90-60 percent downward.

Style C - 40-60 percent upward 60-40 percent downward.

Style D - 60-90 percent upward 40-10 percent downward.

3.15 Efficiency (type II fixtures only). Fixtures shall have the minimum efficiency as indicated in table II when tested in accordance with 4.6.3. Efficiency as used herein is the ratio, in percent, of luminous flux leaving a fixture to the total luminous flux emitted by the lamp or lamps used therein.

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TABLE II.

Style	2 Lamp	4 Lamp
Style A - Direct	60 percent	45 percent
Style B - Semi-direct	60 percent	45 percent
Style C - Direct-indirect	75 percent	70 percent
Style D - Semi-indirect	75 percent	70 percent

3.16 Visual Comfort Probability and Luminance (type II fixtures only). When determined in accordance with the procedures specified in 4.6.4, fixture Visual Comfort Probability (VCP) shall be 70 or more for crosswise and lengthwise orientations. Additionally, when determined in accordance wigh 4.6.2 the ratio of maximum to average luminaire (fixture) luminance (photometric brightness) shall not exceed five to one at angles of 45°, 55°, 65°, 75°, and 85° from nadir in the crosswise and lengthwise planes; and maximum luminaire luminances in the crosswise and lengthwise planes shall not exceed the following values:

Angle Above Nadir (degrees)	Maximum Footlamberts
45	2250
55	1605
65	1125
75	750
85	495

3.17 Preproduction Sample (type II fixtures only). Unless otherwise specified (see 6.1), before production is commenced, a sample fixture of each style, mounting and lamp arrangement, specified in the invitation for bids, shall be made available to the contracting officer or his authorized representative for approval in accordance with paragraphs 4.4, 4.5, and 4.6. Test data indicating that all required tests specified or described in paragraphs 4.5 and 4.6 have been successfully completed must accompany each preproduction sample at the time it is made available to the Government. Furthermore, the test data must be obtained through an independent test laboratory acceptable to the Government. Test data indicating compliance with 4.5.1 through 4.5.5 shall be no older than six months. The requirements of this paragraph shall be applicable on a per contract basis.

3.18 Electrical requirements. Type I fixtures shall be designed to operate on either 110-125-volt or 250-290-volt, 60 Hertz alternating current, as specified (see 3.6 and 6.1). Type II fixtures shall be designed to operate on either 110-125-volt or 250-290-volt, 60 Hertz alternating current; or on 210-236-volt, 50 Hertz alternating current, as specified (see 3.6 and 6.1).

3.19 Electromagnetic interference-free requirements. When specified (see 6.1), type II fixtures shall be designed for suppression of electromagnetic interference such that requirements of MIL-STD-461 are complied with. Testing for electromagnetic interference shall be in accordance with paragraph 4.5.6.

3.20 Workmanship. Finished fixtures shall be clean and conform to the quality and grade of the product established by this specification. All burrs and rough and sharp edges shall be removed. Exposed welds or seams shall be ground smooth. There shall be no flaking, peeling, or chipping of paint or exposure of bare metal. All holes shall be accurately and cleanly punched or drilled. Fixtures shall be free from any defects that affect their appearance and serviceability or any defect that may cause injury.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.1), in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities for any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2.2 Quality conformance inspection. Samples selected in accordance with 4.3.1 and 4.3.2 shall be examined and tested in accordance with 4.4, 4.5.2, 4.5.4, and 4.5.5.

4.2.3 Inspection of preparation for delivery. An inspection for preparation for delivery shall be performed in accordance with paragraph 4.7.

4.3 Sampling for quality conformance inspection.

4.3.1 Sampling for visual examination. Sampling for visual examination shall be performed in accordance with provisions set forth in MIL-STD-105. The Inspection Level shall be level II and, except as noted below, the Acceptable Quality Level (AQL) shall be 2.5 percent for major defects, and 6.5 percent for minor defects, expressed in terms of percent defective.

4.3.2 Sampling for tests. Sampling for tests shall be as specified for each test in paragraphs 4.5.2, 4.5.4, and 4.5.5.

4.3.3 Definition of lot. Lot as used herein shall be the collection of units of product from which samples are to be drawn and inspected to determine compliance with specifications requirements. As regards fixtures, a lot shall consist of units of product of a single type, style, mounting and lamp arrangement (see 1.2). Samples units are to be from the same production run.

4.4 Visual examination. The completely assembled end item shall be examined for, but not limited to, the defects listed in table III. The inspection level and AQL shall be as specified in 4.3.1.

4.4.1 Table IV. Classification of defects.

TABLE III.

Defects	Requirement	Classification	
		Major	Minor
Pattern not uniform	3.2.2	X	
Thickness not as specified	3.2.2	X	
Canopy assembly dimensions not as specified	3.2.2.1		X
Movement less than specified	3.2.2.1		X
Parts missing or not included	3.2.2.2, 3.3	X	
Accessibility not as specified	3.2.2.3	X	
Rough edges, not rounded or bushed	3.3.2, 3.3.3	X	
Plastic and/or glass not as specified	3.4	X	
Lens and panel marking	3.4.3	(see note)	
Lampholders not as specified	3.5	X	
Ballast not as specified	3.6	X	
Parts not as specified	3.7	X	
Fabrication not as specified	3.3, 3.8	X	
Knockouts not as specified	3.8.2, 3.8.3		X
Marking not discernible, not as specified	3.9		X
Finish on reflecting surfaces not as specified	3.10.1	X	
Finish on non-reflecting surfaces not as specified	3.10.2		X
Parts not interchangeable	3.13	X	

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Note: Lens closure and panel (if used) identification marking (see 3.4.3) shall be compared with identification information on photometric test report (see 4.6.5). If information on test report, excepting date of manufacture, does not correspond with identification marking on lens closure and panel, the entire fixture lot shall be rejected. Sampling shall be as specified in 4.3.1, no defects shall be allowed.

4.5 Tests.

4.5.1 Plastic. Test results indicating compliance with ASTM D788 shall be provided by plastic manufacturer for each lot of plastic used in the manufacture of fixtures covered herein.

4.5.1.1 Rate of burning. The rate of burning shall be as indicated in 3.4.2 when tested in accordance with Fed. Test Std. No. 406 Method 2021.

4.5.1.2 Test frequency. The tests in 4.5.1.1 shall be performed once for each class, grade or type of plastic used in the manufacture of the fixtures furnished under the specification.

4.5.2 Salt spray test. The fixture shall withstand the salt spray test as specified in Method 811.1 of Fed. Test Method No. 151 for a period of 48 hours (see 3.11). Only one sample per lot shall be tested. Failure shall be cause for rejection of the lot.

4.5.3 Exposure test. At least five sample plates shall be exposed for one hundred hours to the radiation of a glass enclosed carbon arc lamp, such as a Fade-O-Meter, operating on a current of 13 ± .5 amperes and electrical potential, at the arc, of 140-volts. Each sample plate shall be flat, 4 inches wide by 6 inches in length, and must be constructed of the same material and have the same finish as the fixture reflecting surface it represents. Each sample plate shall be placed ten inches from the arc and the lamps so ventilated that the temperature at the sample does not exceed 105°F. After exposure in the above manner, determine the average reflectance by measuring the reflectance of each sample with a reflectometer, or instrument providing equivalent results, and averaging the cumulative reflectance measurements. After obtaining average reflectance, determine hiding of black and white substrate in accordance with method 4121 of Fed. Test Method Std. No. 141. Ascertain compliance with 3.10.1 and 3.10.1.1. This test need only be conducted for preproduction sample fixture(s) when a preproduction sample(s) is required, or only for the first lot of fixtures of each style, mounting and lamp arrangement specified in the invitation for bids when a preproduction sample(s) is not required. Decrease in average reflectance of more than 4 percent, signs of loss of adhesion on any sample, or failure of more than 40 percent of samples to comply with the hiding of black and white substrate requirement shall be construed as test failure and shall be cause for lot rejection.

4.5.4 Average reflectance. Average reflectance shall be determined by averaging the reflectance values taken at five randomly selected points on reflector surface. The average value so obtained shall be at least 96 percent of the value given in independent test lab data (see 3.10.1.1 and 4.6.5). Acceptable Quality Level (AQL) shall be 2.5, expressed in percent defective, in accordance with MIL-STD-105. Inspection Level shall be level II.

4.5.5 Rigidity test. Fixture shall be placed with bottom closure face down on a rigid horizontal support surface. The fixture shall be lifted by application of a vertical force at one corner until such time as three corners are clear of the support surface; the fourth corner (diagonally opposite lift corner) only remaining on the support. When in the correct test position the fixture shall be supported only at the lift corner and the corner diagonally opposite lift corner. The test position shall be maintained for at least one minute whereupon compliance with requirements of 3.12 shall be determined. The above test shall be repeated until the lift force has been applied, in turn, at each of the four fixture corners. Four samples per lot shall be tested. Failure of any sample shall be cause for rejection of the lot.

4.5.6 Electromagnetic interference. When specified in the invitation for bids (see 6.1 electromagnetic interference suppression tests shall be conducted in accordance with MIL-STD-461, and 462. Refer to MIL-STD-463, as required, for definitions and system of units used in electromagnetic interference technology. The sample for this test shall be one complete fixture of each style, mounting and lamp arrangement specified in the invitation for bids. Testing must be accomplished by an independent laboratory acceptable to the Government. This test need only be conducted for preproduction sample fixture(s) when a preproduction sample(s) is required, or only for the first lot of fixtures. Failure shall be cause for rejection of the lot.

4.6 Light distribution, luminance, efficiency and visual comfort probability (type II fixtures only). All tests described in paragraph 4.6 shall be conducted on one fixture of each style, mounting and lamp arrangement specified in the Invitation for Bids. Additionally, all tests must be conducted by an independent laboratory acceptable to the Government.

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4.6.1 Light distribution. In accordance with procedures outlined on pages 4-16, 4-17 and 4-18 of the IES Lighting Handbook, Fifth Edition (1972) under the general section titled "Photometry of Luminaires" and subsections titled "General Lighting Luminaires (Candlepower Distribution and Luminance (Photometric Brightness))";"General Lighting Luminaires (Total Lumen Output)"; and "Calculations of Luminaire Lumens From Candlepower Data", perform the following:

(a) Obtain fixture candlepower distribution data in at least 3 vertical planes; a vertical plane parallel to the fixture's lamps, a vertical plane perpindicular to the fixture's lamps, and an intermediate vertical plane of 45 degrees; the vertical planes being with respect to the lamps longitudional axis. Candlepower readings shall be taken at intervals of 10 degrees or less in the center of each interval (zone).

(b) From the above candlepower data calculate the luminous flux values for each zone from nadir to 180 degrees above nadir.

(c) Compute the percentage of light distribution utilizing the luminous flux values calculated above and the following relations:

$$\text{Percent Upward Light Distribution} = \frac{\text{Lumen Output } (90^\circ \text{ to } 180^\circ \text{ zones}) \times 100}{\text{Lumen Output } (0^\circ \text{ to } 180^\circ \text{ zones})}$$

$$\text{Percent Downward Light Distribution} = \frac{\text{Lumen Output } (0^\circ \text{ to } 90^\circ \text{ zones}) \times 100}{\text{Lumen Output } (0^\circ \text{ to } 180^\circ \text{ zones})}$$

4.6.2 Luminance. In accordance with the procedures cited on pages 4-17, 4-18 and 4-25 of the above mentioned IES Handbook, under the subsections titled "General Lighting Luminaires (Candlepower Distribution and Luminance (Photometric Brightness))" and "Luminance (Photometric Brightness) Measurements," and procedures outlined in Journal of IES, Volume I, No. 2, January, 1972, beginning on page 181, titled "Determination of Average Luminance of Luminairies," determine the average and maximum luminance for the angles and planes cited in paragraph 3.16.

4.6.3 Efficiency. Fixture efficiency shall be determined in accordance with either of the procedures outlined on page 4-18 of the above mentioned IES Handbook under the subsection titled "General Lighting Luminaires (Total Lumen Output)."

4.6.4 Visual Comfort Probability (VCP). Visual Comfort Probability (VCP) shall be determined in accordance with the procedures outlined in the Illuminating Engineering Society's RQQ Report No. 2 (1972) - Outline of a Standard Procedure For Computing Visual Comfort Ratings For Interior Lighting." VCP calculations shall be based on a room size of 60 feet x 60 feet, fixture mounting height of 10 feet above floor level, and the standard conditions cited in above RQQ Report No. 2 (1972).

4.6.5 Photometric tests and test data. Any changes made to the fixture which would in any way affect 3.14, 3.15 or 3.16 shall require the performance of new photometric tests in accordance with 4.6. Data as furnished in compliance with 3.14, 3.15, and 3.6 shall contain average fixture reflectance value, lens closure and panel (if used) identification as cited in 3.4.3, and shall be no older than five years. Date (month and year) of testing shall be indicated on all test reports. Average fixture reflectance shall be determined as specified in 4.5.4. Additionally, photometric tests shall be based on the assumption that each lamp contained within the fixture produces 3100 lumens rated initial output. If lamps used for tests do not produce initial rated output, candlepower measurements and data shall be adjusted so that they equal what lamps would produce at rated initial output. See 1.2 for lamp designations.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table IV. For examination of interior packaging, the sample unit shall be one shipping container prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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4.1.2 No company supplying any [illegible] to the contractor

TABLE IV. Classification of preparation for delivery

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of flaps, loose strapping, inadequate stapling. Distortion of the container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A. The fixtures shall be packaged in accordance with MIL-P-116, submethod 1C-2.

5.1.2 Level B.

5.1.2.1 Type I fixtures. Type I fixtures shall be packaged as applicable in paragraphs 5.1.2.1.1, 5.1.2.1.2, and 5.1.2.1.3.

5.1.2.1.1 Louver. The louver assembly shall be inserted in the unit and secured in place with pressure sensitive adhesive tape applied either around the holders and ends of rods inserted therein or by encircling the center transformer mounting frame and the center plates of the diffuser assembly at points approximately six inches from each end.

5.1.2.1.2 Installation hardware. All hardware required for installation shall be adequately cushioned, then wrapped in a flexible corrugated fiberboard wrap or inserted in a padded shipping bag with ends securely taped to prevent loss of content. Size of wrapped or bagged package shall allow placement between center transformer frame and one side of the fixture when packing.

5.1.2.1.3 Stems (when furnished separately). Ten stems with nuts assembled on each stem shall be packaged in a paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676 or to PPP-B-636. Each box shall be secured to prevent accidental opening. Alternatively ten stems shall be wrapped in kraft paper and secured by means of pressure sensitive reinforced adhesive tape conforming to PPP-T-97 or by non-metallic strapping conforming to PPP-S-760 grade A.

5.1.2.2 Type II fixtures. The fixtures and accessories shall be individually packaged utilitzing necessary blocking, bracing and tape to prevent damage to the item during shipment and storage. The manufacturer's standard packaging procedures shall be acceptable providing that these prerequisities are met.

5.1.3 Level C. The fixtures and accessories shall be packaged to insure protection during shipment and shall provide for safe delivery to their destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Complete fixtures. Each complete fixture packaged, as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-621, class 2; PPP-B-601, overseas type; PPP-B-585, class 3, or to PPP-B-640, class 2. The box shall be closed and reinforced in accordance with the appendix to the applicable box specification.

5.2.1.2 Stems (when furnished separately). The stems packaged, as specified in 5.1.2.1.3 shall be packed in the quantities indicated in table V, in a close-fitting box conforming to PPP-B-636, class weather-resistant, style RSCL. Closure of the individual boxes shall be in accordance with the appendix to PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Complete fixtures. Each complete fixture, cushioned and protected as specified in 5.1.2, shall be packed in close-fitting box conforming to PPP-B-636, class domestic, style FTC, grade 275, minimum. The box shall be long enough to permit a double-face corrugated spring pad at each end to center the fixture in the container. Closure of the box shall be in accordance with method II of the appendix to PPP-B-636.

5.2.2.2 Stems (when furnished separately). The stems packaged, as specified in 5.1.2.1.3, in the quantities indicated in table V, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, style RSCL. Closure of the individual boxes shall be in accordance with method II of the appendix to PPP-B-636.

TABLE V.

Size	Stems Per Unit Package	Unit Packages Per Shipper
Inch		Stems
4	10	20 (200)
7	10	20 (200)
30	10	6 (60)
24	10	6 (60)
18	10	6 (60)
12	10	20 (200)
48	10	4 (40)

5.2.3 Level C. The fixtures, packaged as specified in 5.1, shall be packed to comply with the Uniform Freight Classification rules or the National Motor Freight Classification rules, as applicable.

5.3 Palletization. When specified (see 6.1), fixtures packed as specified in 5.2, shall be arranged on pallets in quantities of the option of the supplier, provided that the overall dimensions of the loaded pallet do not excede 6 feet in length, 5 feet in width and 5 feet in height. Each succeeding tier of fixtures shall be quarter-turned on the layer beneath. The design of the pallet shall conform to type II of NN-P-71, except that the length and width dimensional requirements shall be waived and the spcaing between deck boards shall be no more than 3 inches. The load shall be secured to the pallet with a minimum of two straps perpendicular to the length to the pallet width. Strapping used shall conform to QQ-S-781, type I, class A, size .500 x .020 inches or to PPP-S-760, type II or III, size .500 x .030 inches. The top outer packs shall be provided with edge protectors under the straps to prevent crushing of the container edges when fiberboard boxes are utilized.

5.4 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.4.1 or 5.4.2, as specified (see 6.1).

5.4.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

5.4.3 Special marking. In addition to the marking required by 5.4.1 or 5.4.2 the front, back and end panels of each shipping container shall be marked in each upper corner with the word "UP" and an arrow pointing to the top of the container. The words "UP" shall be printed in characters a minimum of 1 inch high. Additionally, the following shall be marked where indicated in characters a minimum of 1 inch high:

Top panel of box: TOP

Front and back panels of box: THIS SIDE UP FRAGILE-HANDLE WITH CARE

FRAGILE
HANDLE WITH CARE

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.

(b) Type, style, mounting, and lamp arrangement desired (1.2).

(c) Type of bottom closure, if other than specified (3.2.2).

(d) Length of suspension stems, if other than specified (3.2.2.1).

(e) Marking, if other than specified (3.9).

(f) Reflecting surface finish, if other than specified (3.10.1).

(g) Nonreflecting surface finish, if other than specified (3.10.2).

(h) Salt spray test, if required (3.11).

(i) Preproduction sample, if not required (3.17).

(j) Fixture voltage and frequency rating required (3.18).

(k) Electromagnetic interference testing, if required (3.19).

(l) Responsibility for inspection, if other than specified (4.1).

(m) Level of packaging and packing required (5.1 and 5.2).

(n) Palletization, if required (see 5.3).

(o) Special markings required by the contract or order, if required (see 5.4.1 and 5.4.2).

6.2 Other fixture types. Surface ceiling fixtures, previously covered in interim versions of this specification, are now covered in the latest version of specification W-F-1662. Refer also to latest version of specification W-F-1662 for recessed fixtures and to latest version of W-F-1234 for industrial fixtures.

Preparing Activity:

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION –	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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